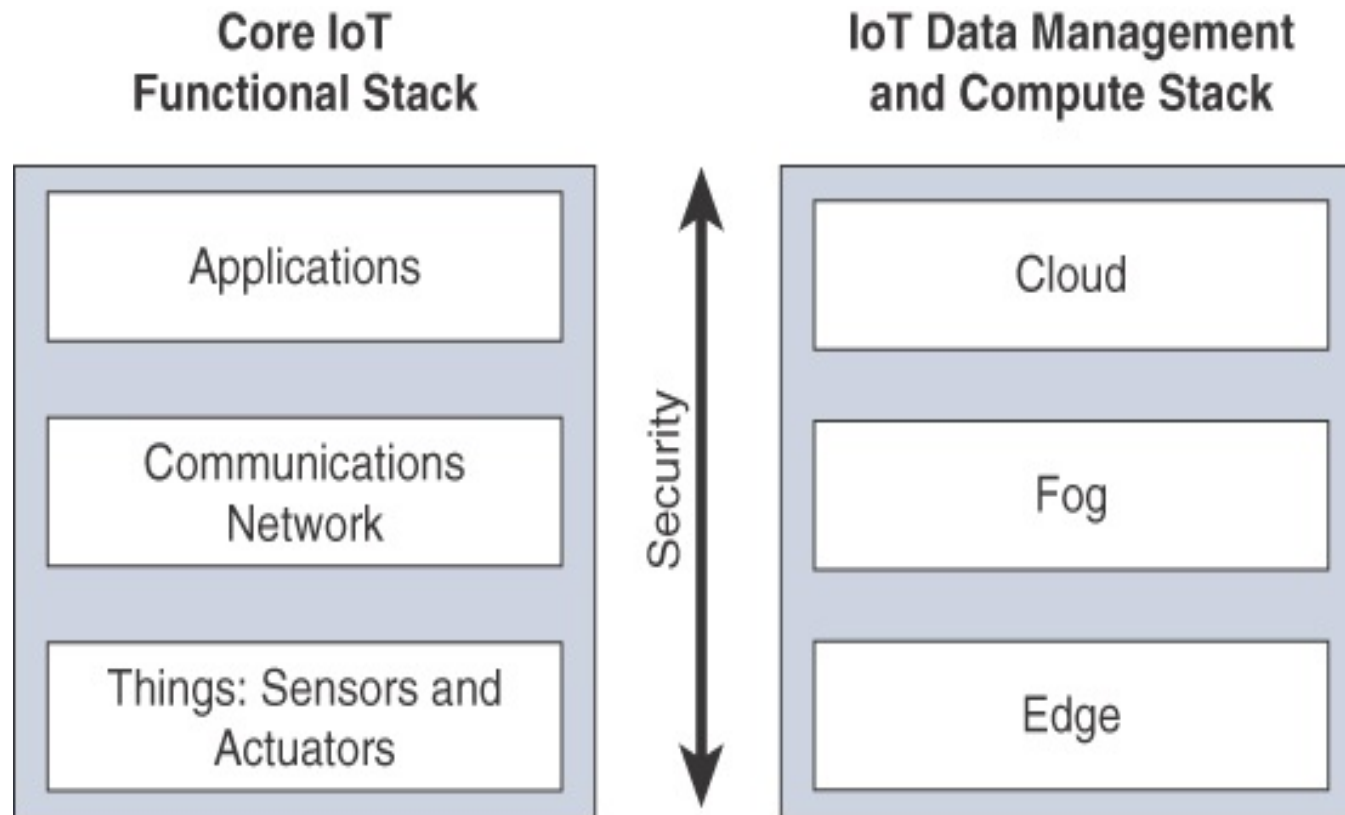


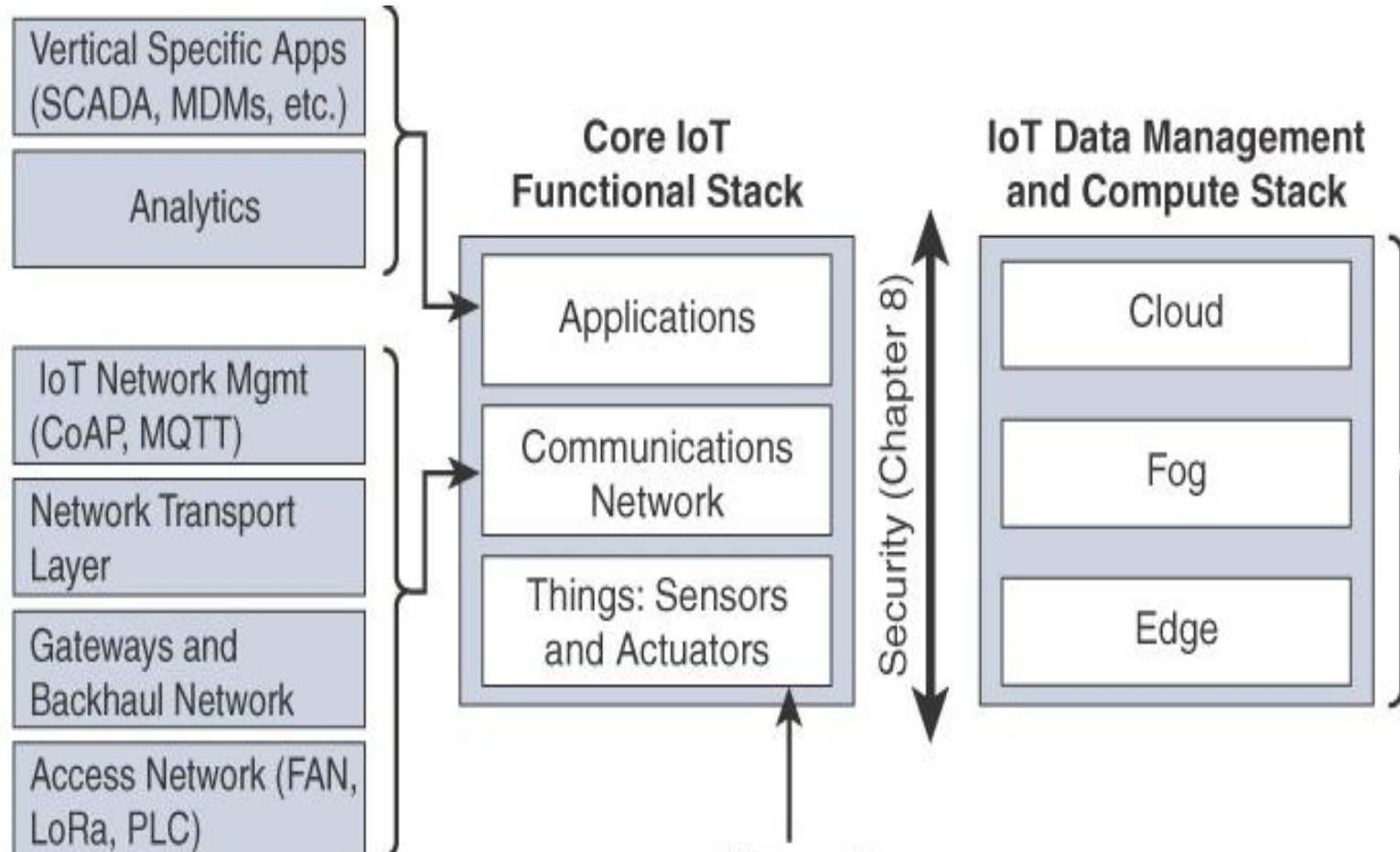
Introduction to Internet of Things

Lecture: 15 & 16

Simplified IoT Architecture



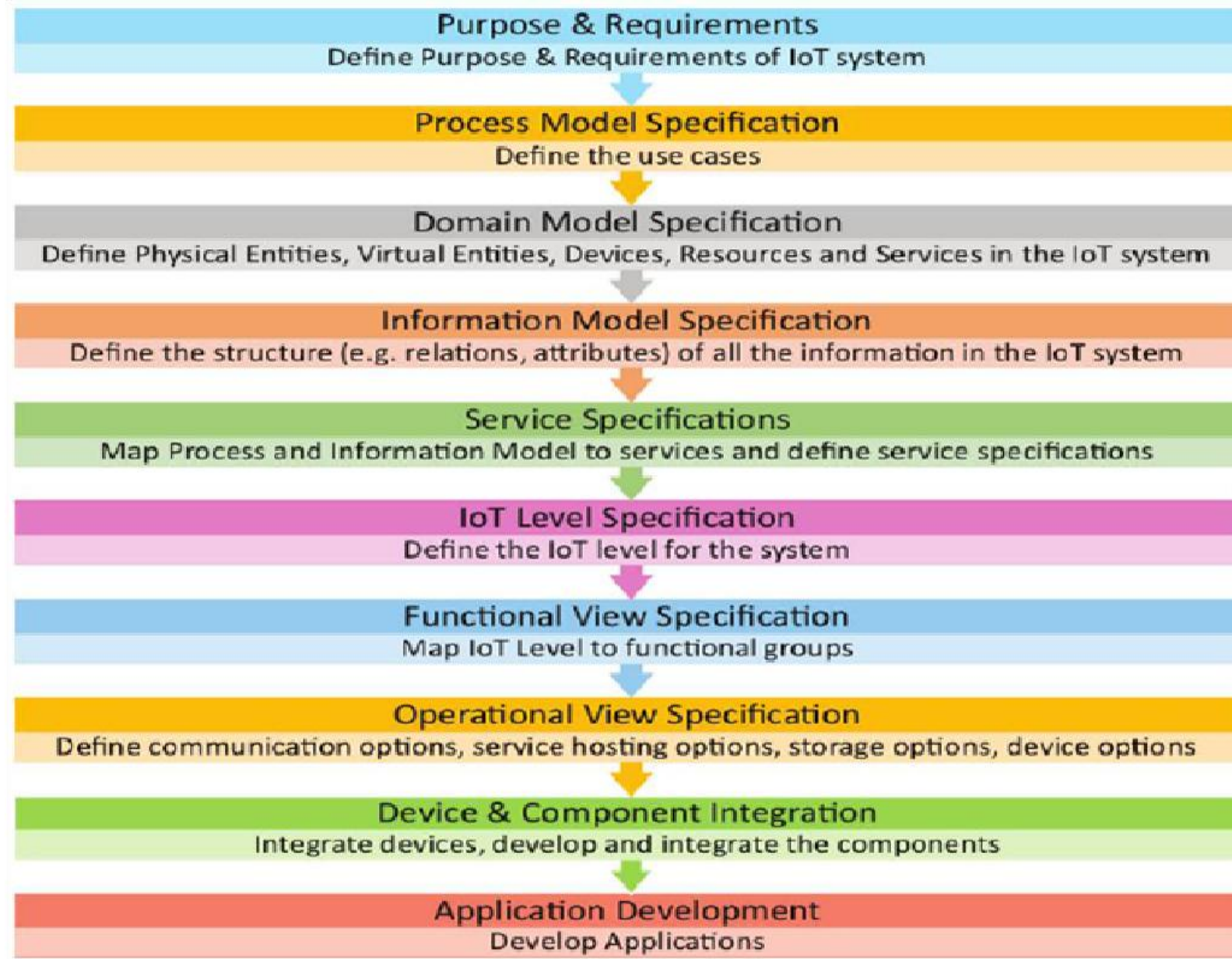
Expanded View



IoT Design Methodology

- ▶ IoT systems designed with the proposed methodology have reduced design, testing and maintenance time, better interoperability. and reduced complexity.





1. Purpose and Requirements Specification

In this first step, define the purpose, behavior and requirements of the system.

Requirements can be:

- Data collection requirements
- Data analysis requirements
- System management requirements
- Data privacy and Security requirements
- User interface requirements

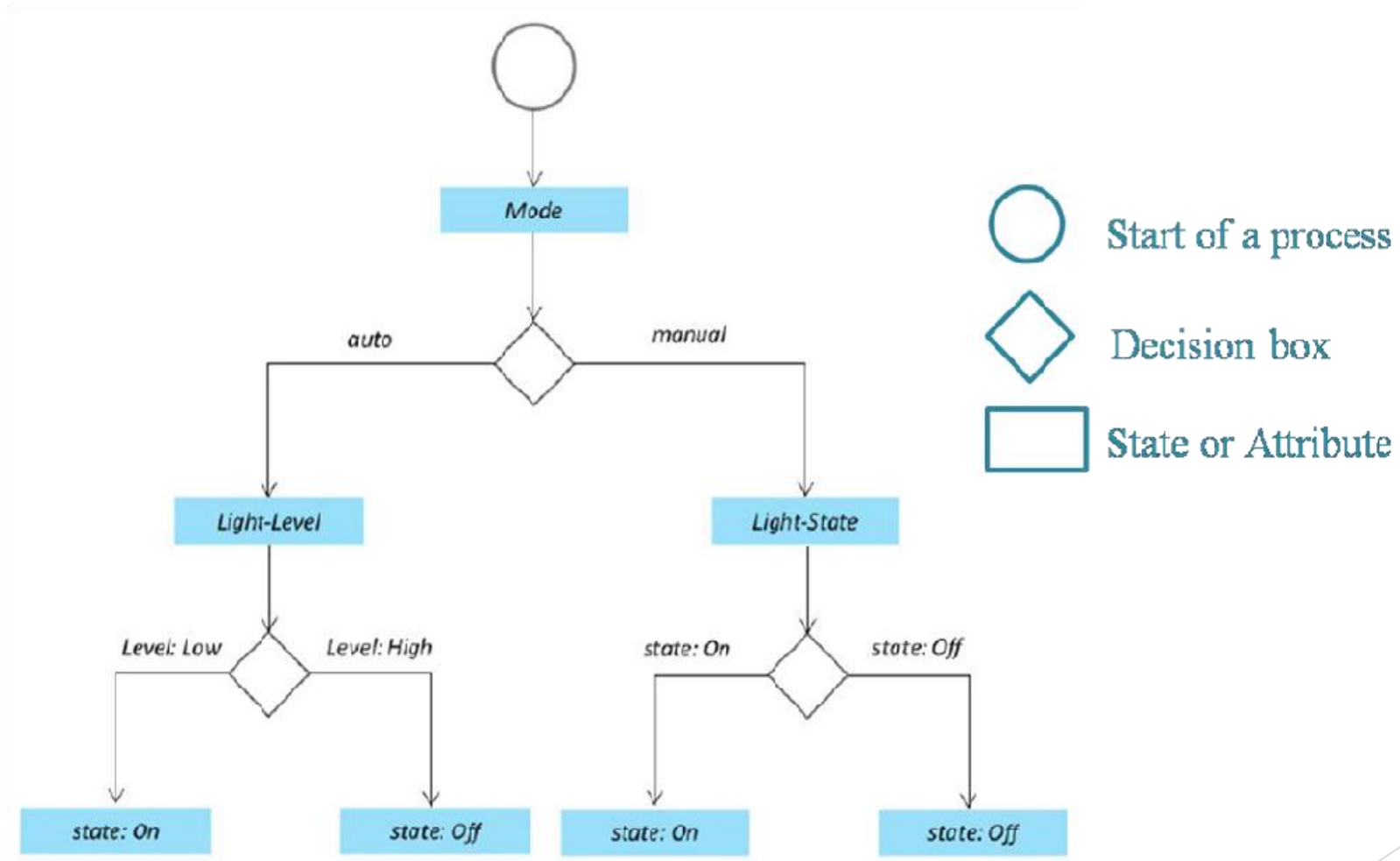
Case study: Home Automation System

Purpose	A home automation system that allows controlling the lights remotely using a web application.
Behavior	Home automation system should support two modes: auto and manual Auto: System measures the light level in the room and switches on the light when it gets dark. Manual: Allows manually and remotely switching lights on and off.
System Management	System should provide remote monitoring and control functions.
Data Analysis	System should perform local analysis of the data.
Application Deployment	Application should be deployed locally on the device, but should be accessible remotely.
Security	Should provide basic user authentication capability.

2. Process Specification

- ▶ In this step, the use cases of the IoT system are formally described based on and derived from the purpose and requirement specifications. It defines how the system will be used.

Use Case: Home Automation System



3. Domain Model Specification

- ▶ The domain model provides an abstract representation of the concepts, entities and objects in the IoT system to be designed. Domain model defines the attributes of the objects and relationships between objects. The domain model is independent of any specific technology or platform.

- **Physical Entity:** The discrete and identifiable entity in the physical environment.

IoT system provides information about the physical entity (using sensors) or performs actuation upon the physical entity (switching on a light).

- **Virtual Entity:** Virtual entity is a representation of the physical entity in the digital world.

For every physical entity, there is a virtual entity in the domain model.

- **Device:** Devices provide a medium for interactions between physical entities and virtual entities.

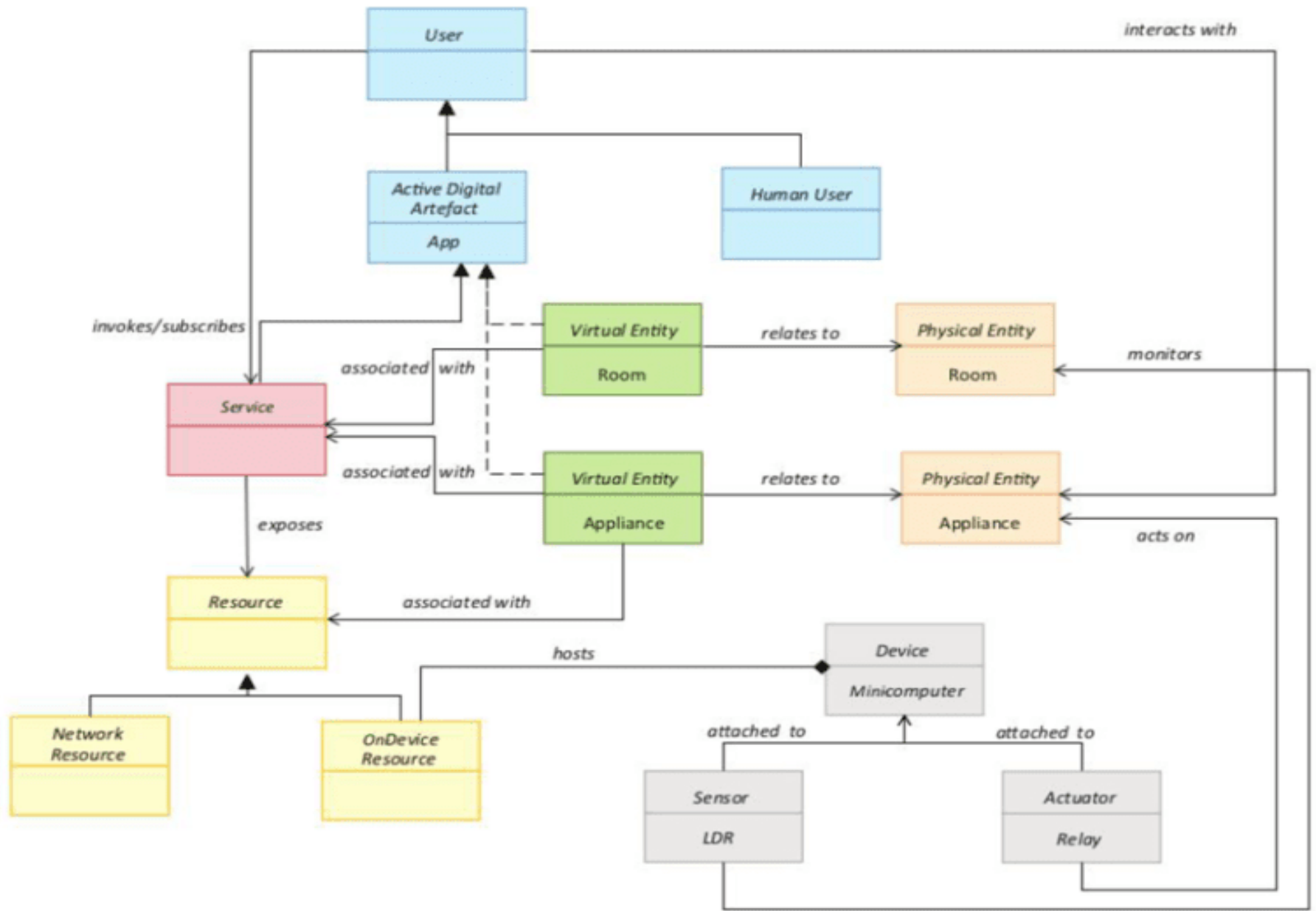
Devices are used to gather information from or perform actuation on physical entities.

- **Resource:** Resources are software components which can be either on-device or network-resources.

On-device resources are hosted on the device and provide sensing or actuation (eg: operating system)

Network-resources include software components that are available in the network (eg: database)

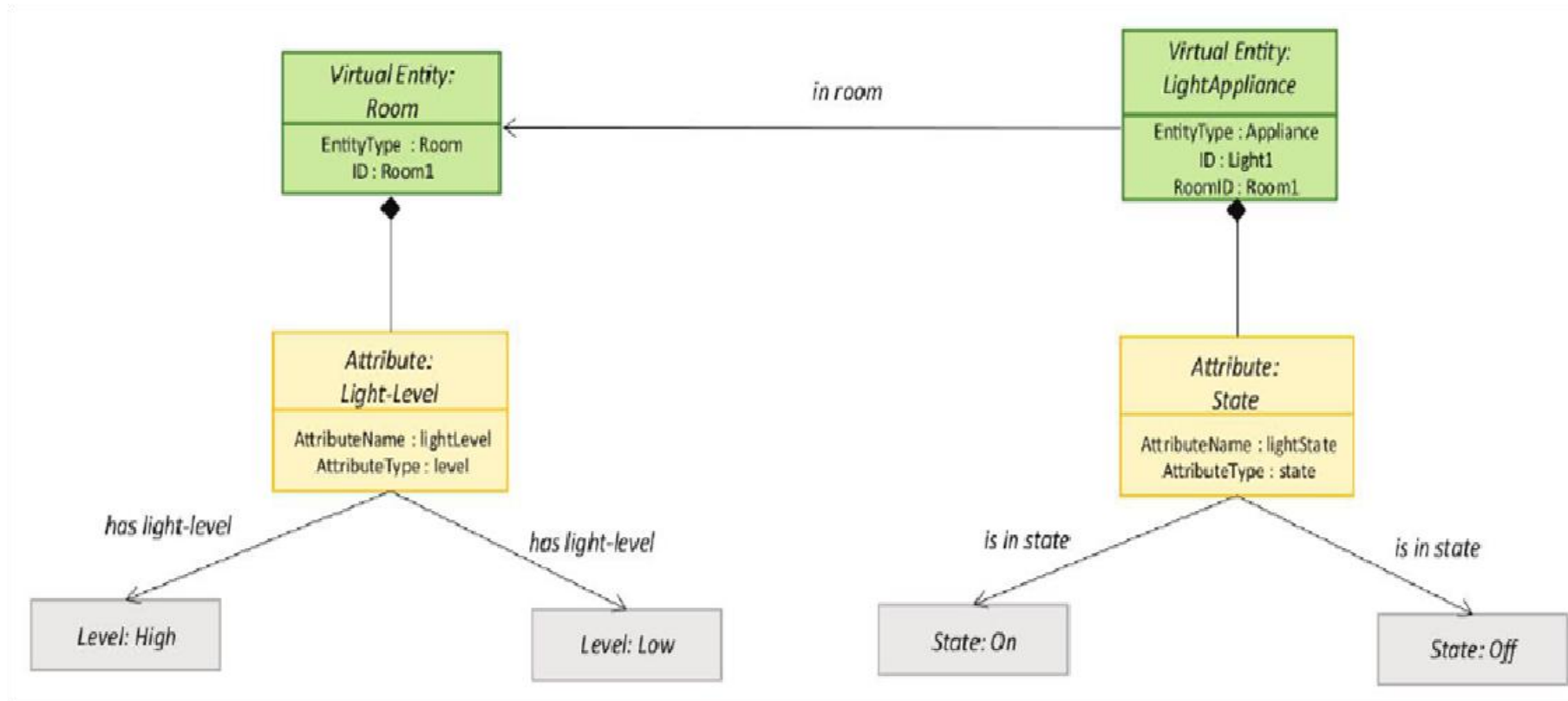
- **Service:** Services provide an interface for interacting with the physical entity. Services access resources to perform operations on physical entities.



4. Information Model Specification

- ▶ Information model defines the structure of all the information in the IoT system. It defines attributes of virtual entities, relations, etc. However, it does not describe how the information is stored and represented. To define the information model, we first list the virtual entities. Later more details like attributes and relationships are added.

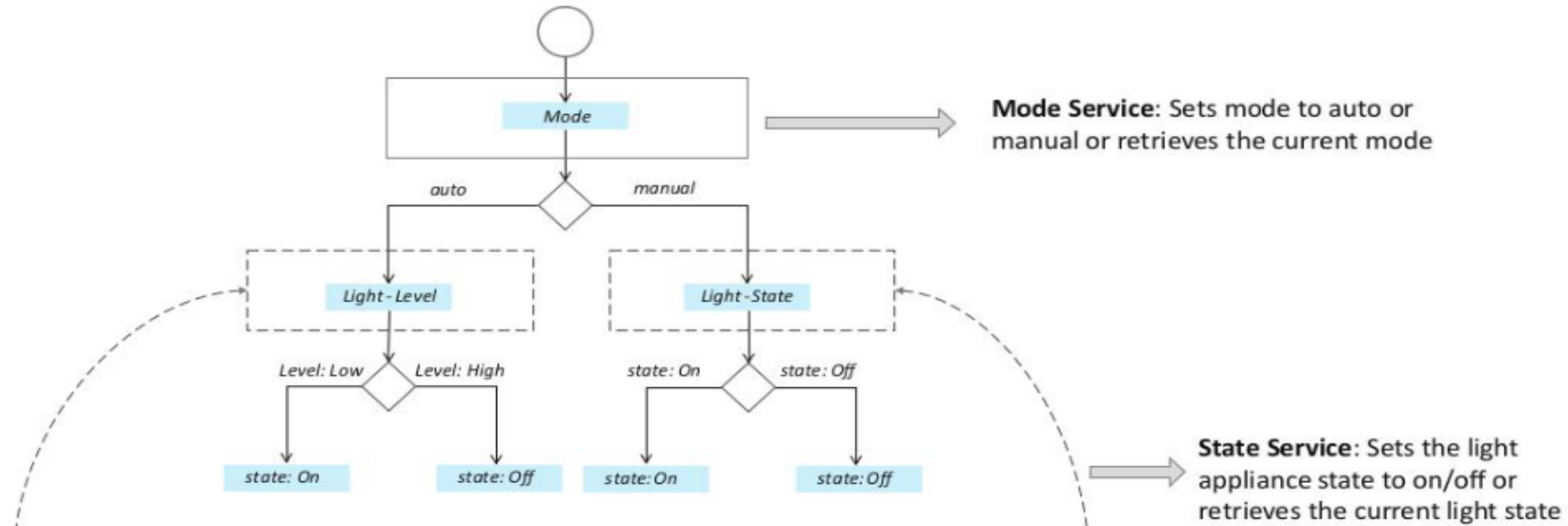
For the Use Case



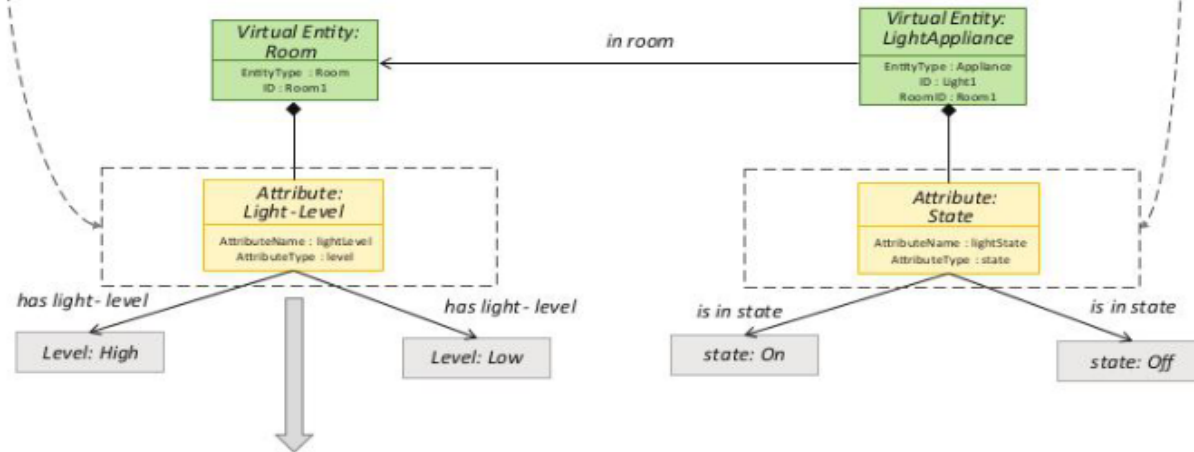
5. Service Specifications

- ▶ The service specification defines the following:
- ▶ Services in the system
- ▶ Service types
- ▶ Service inputs/output
- ▶ Service endpoints
- ▶ Service schedules
- ▶ Service preconditions
- ▶ Service effects

Process Specification

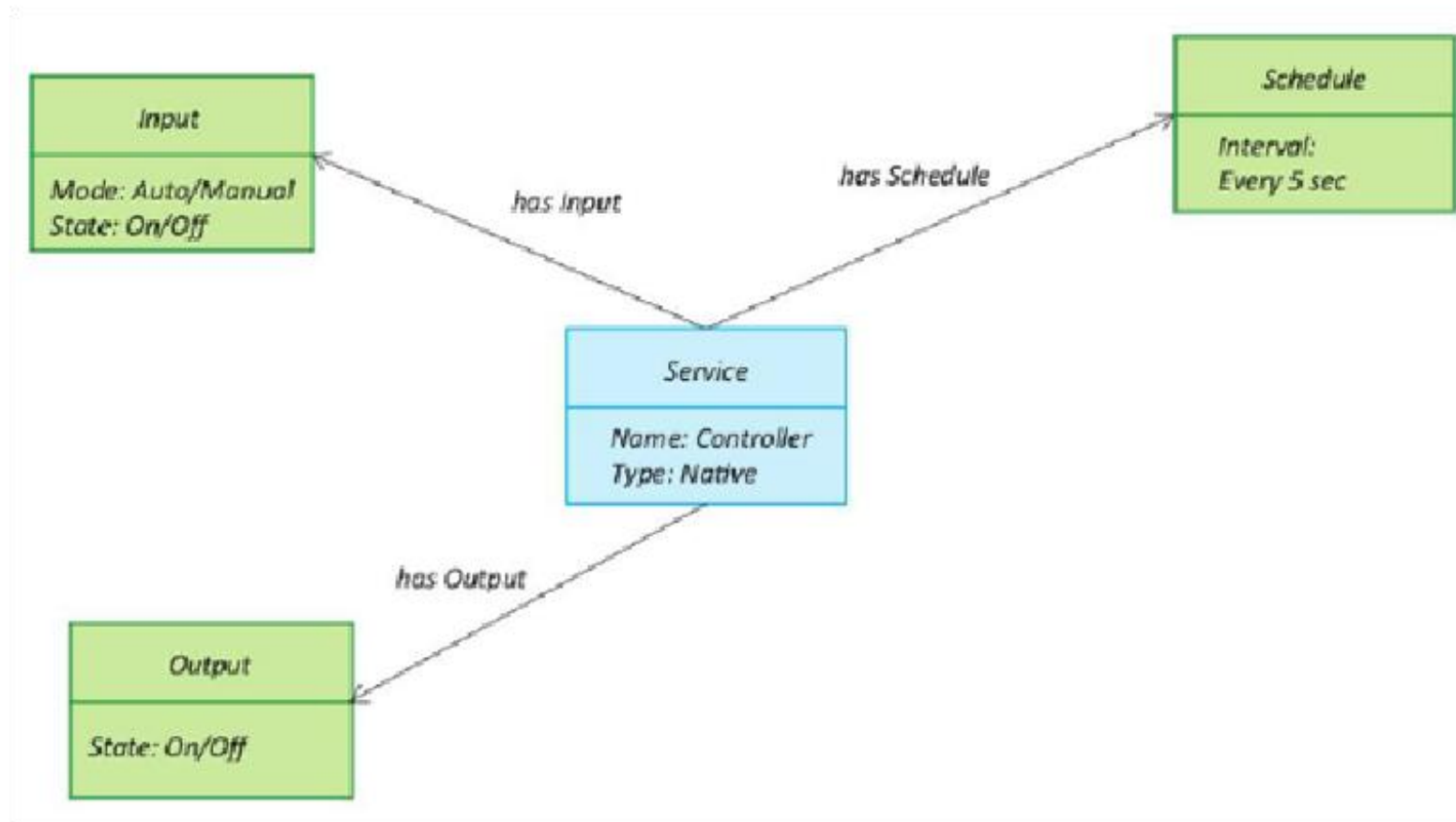


Information Model

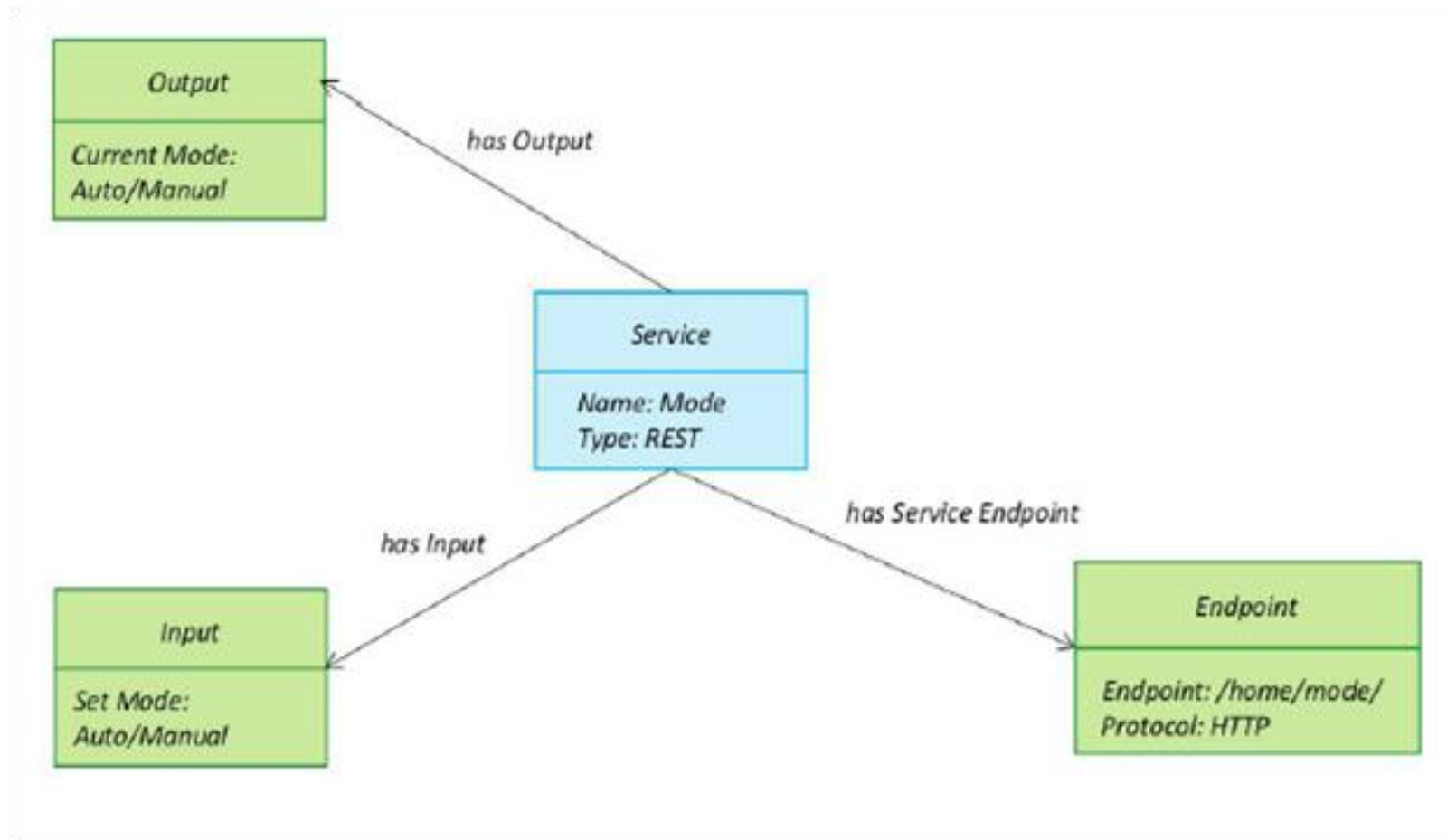


Controller Service: In auto mode, the controller service monitors the light level and switches the light on/off and updates the status in the status database. In manual mode, the controller service, retrieves the current state from the database and switches the light on/off.

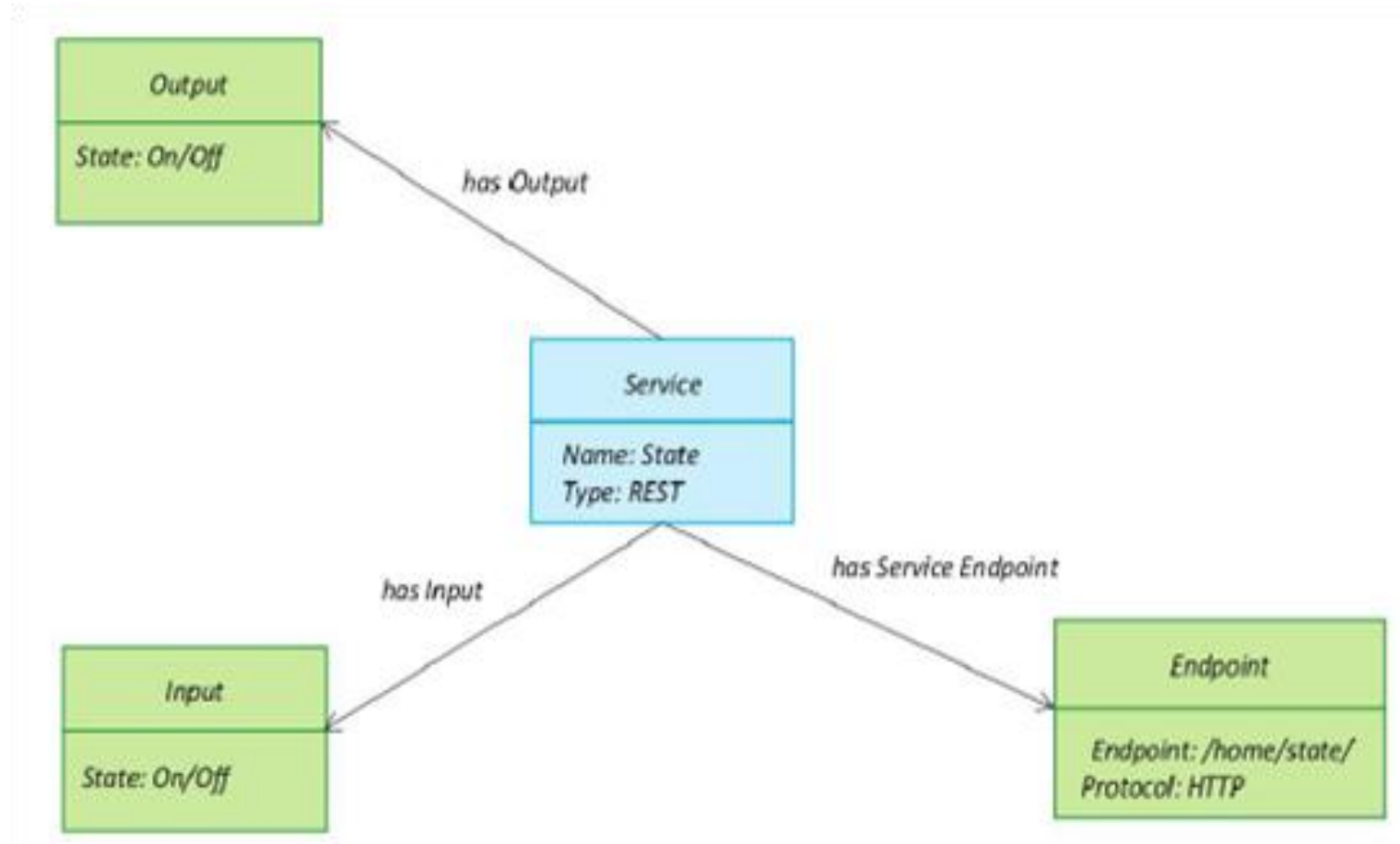
Controller Service of the Use Case



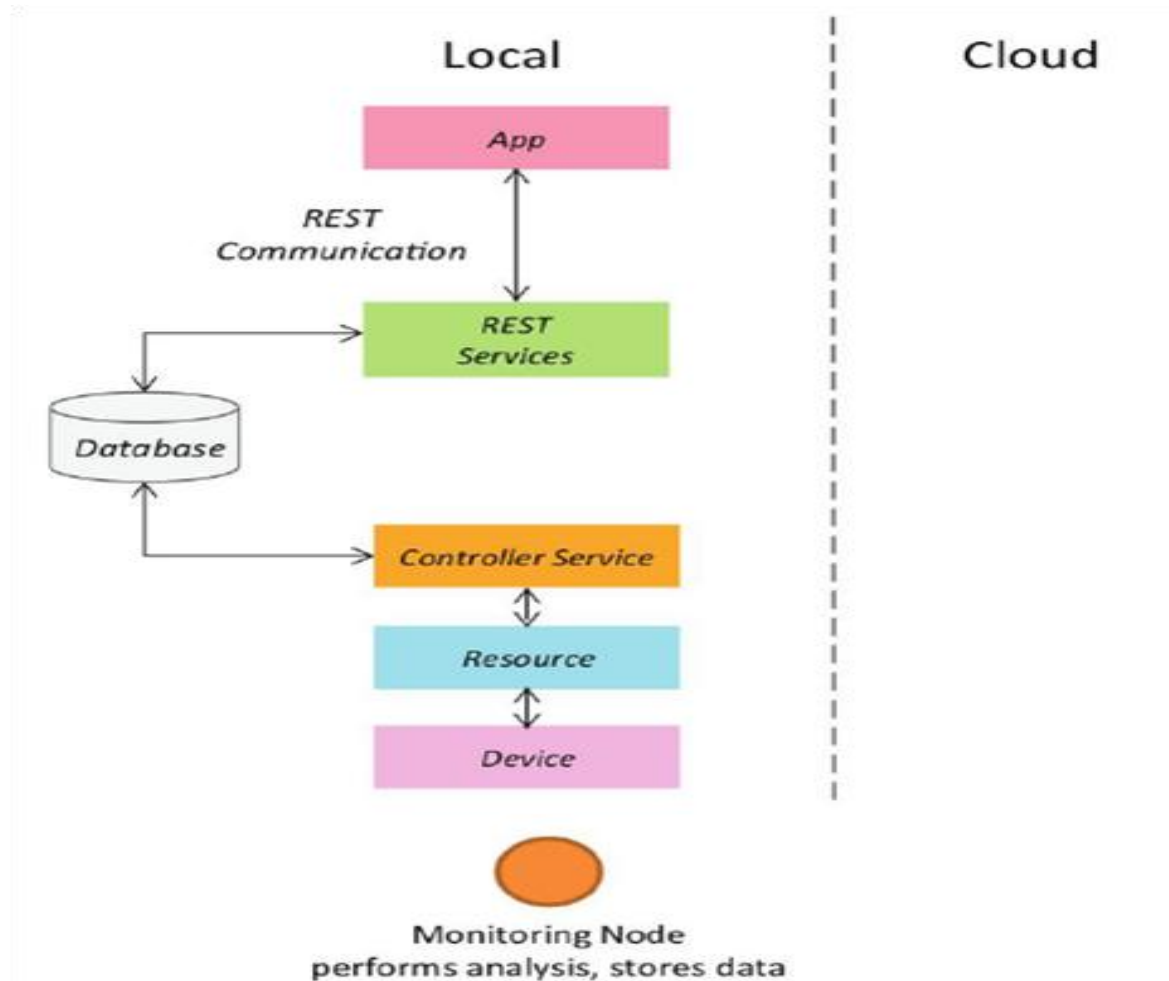
Mode Service of the Use Case



State Service of the Use Case

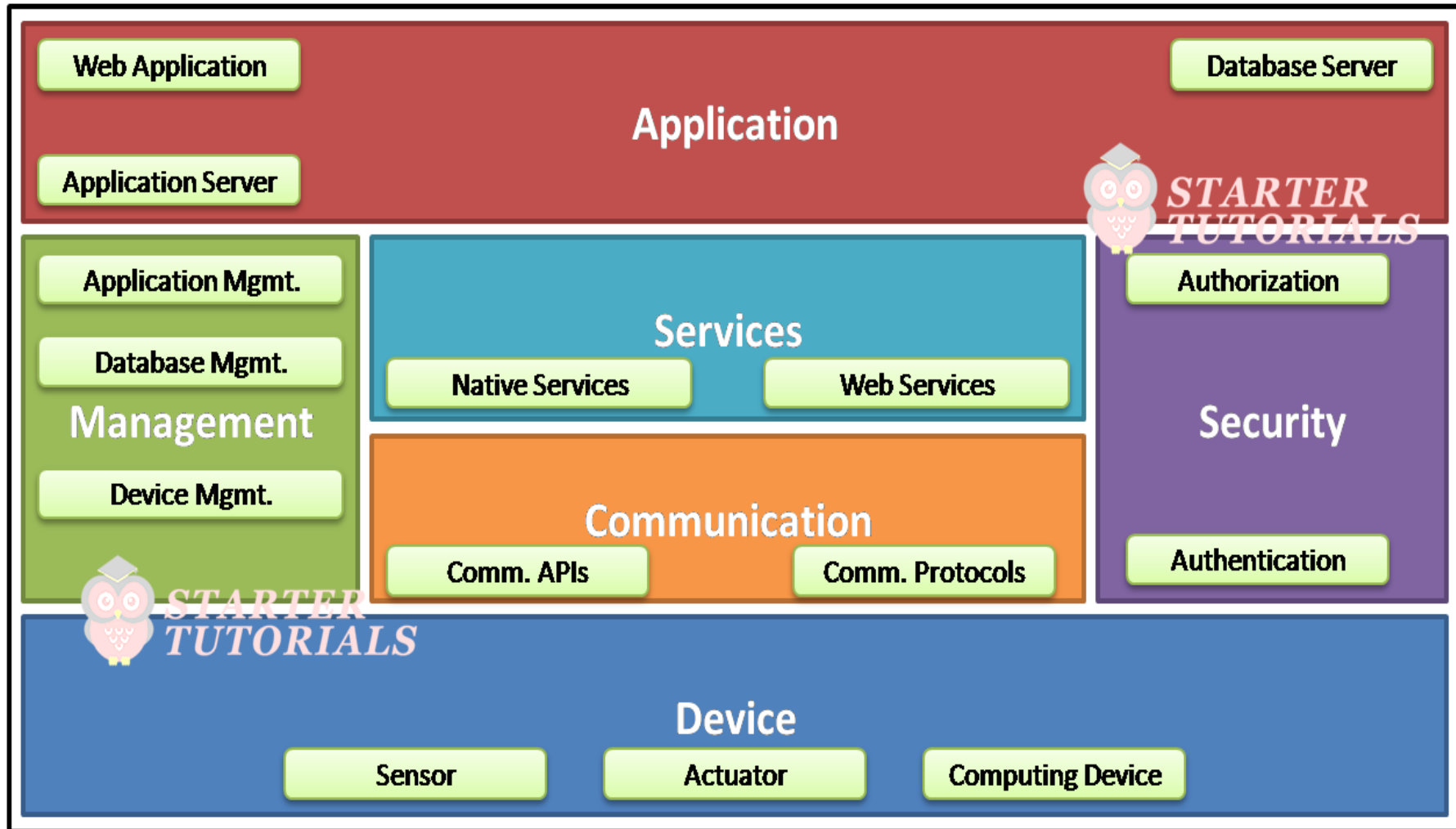


6. IoT Level Specification

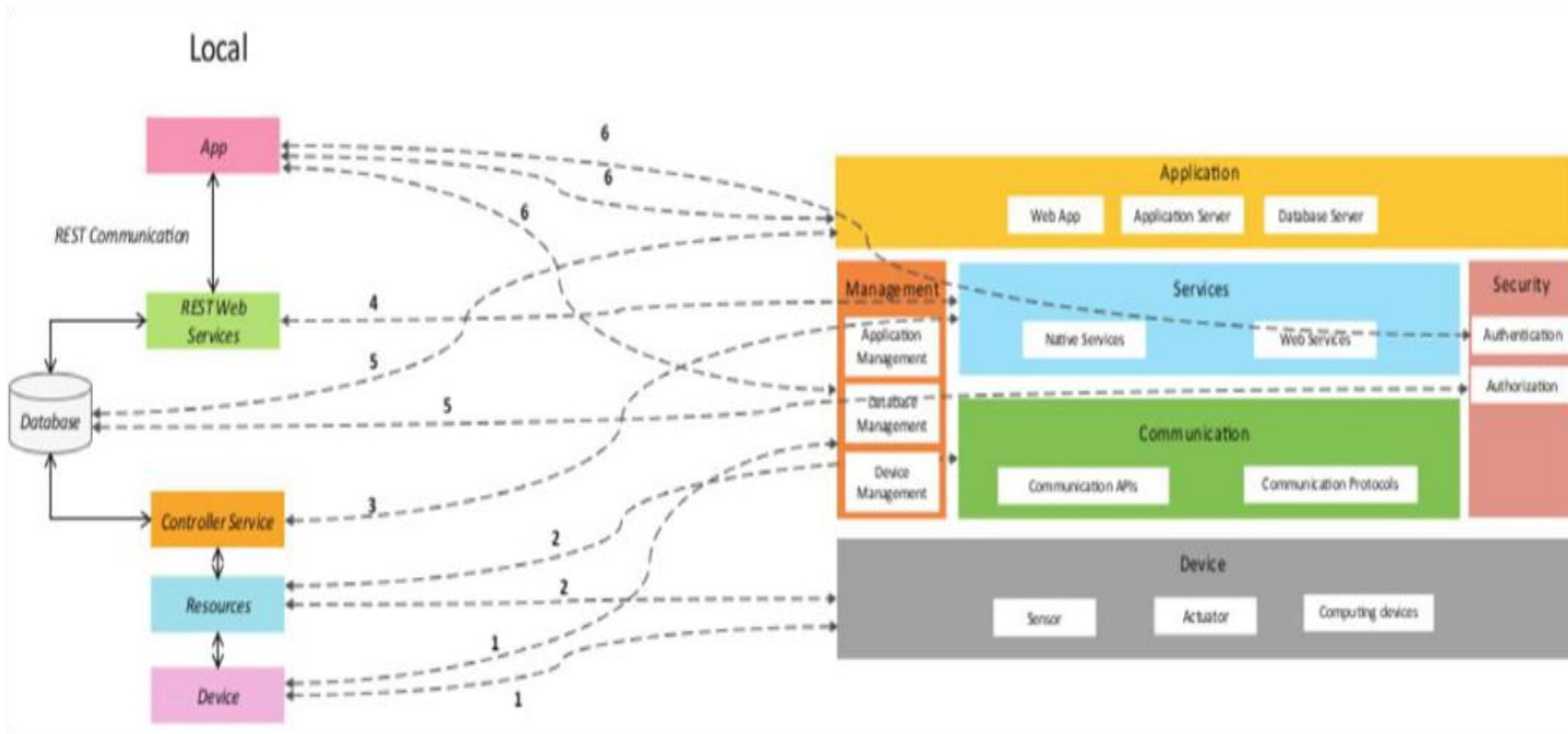


7. Functional View Specification

- ▶ The Functional View (FV) defines the functions of the IoT systems grouped into various Functional Groups (FGs). Each functional group provides functionalities for interacting with concepts in the domain model and information.
- ▶ The Functional Groups included in a Functional View include:
 - Device
 - Communication
 - Services
 - Management
 - Security
 - Application.



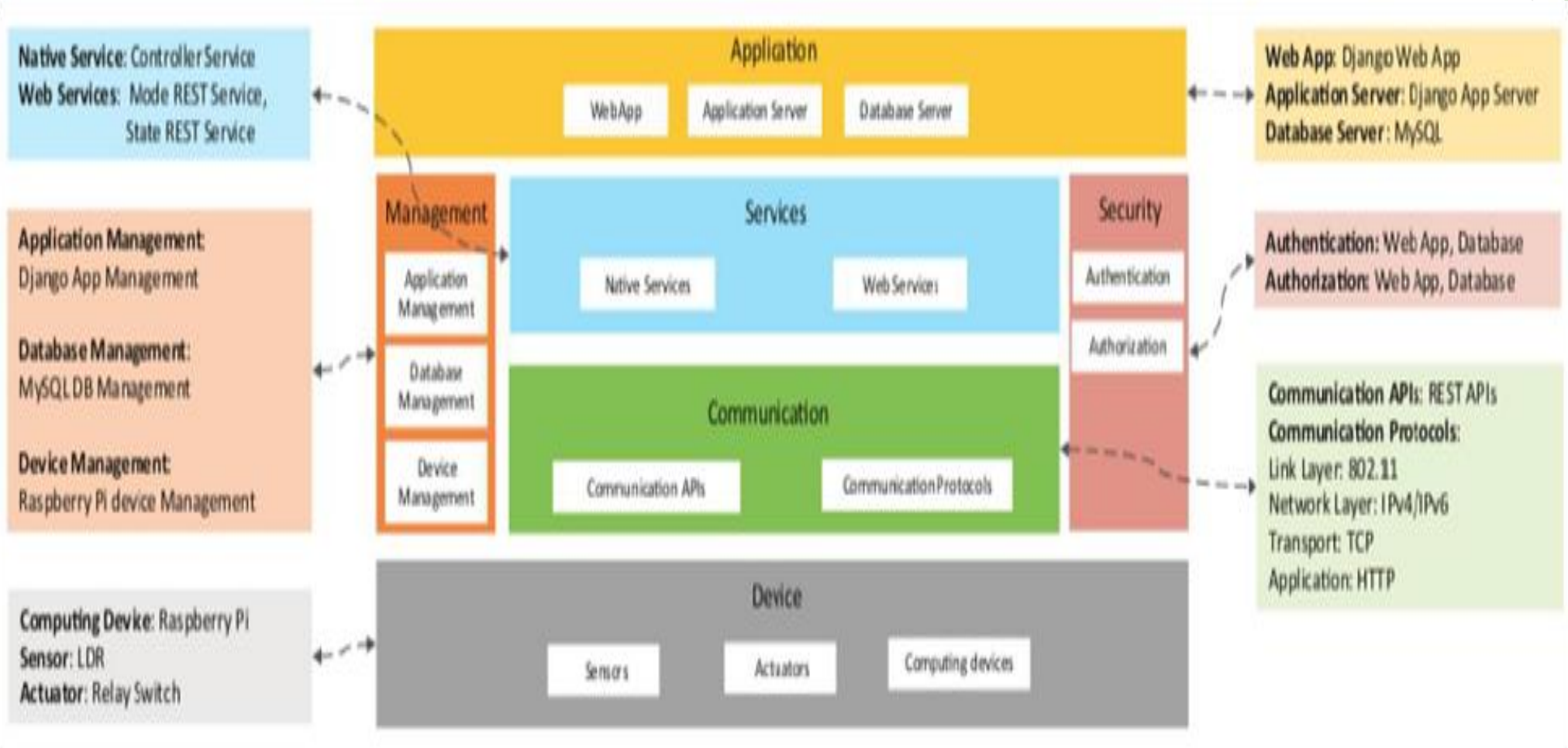
Mapping Deployment level to Functional Groups for the Use Case



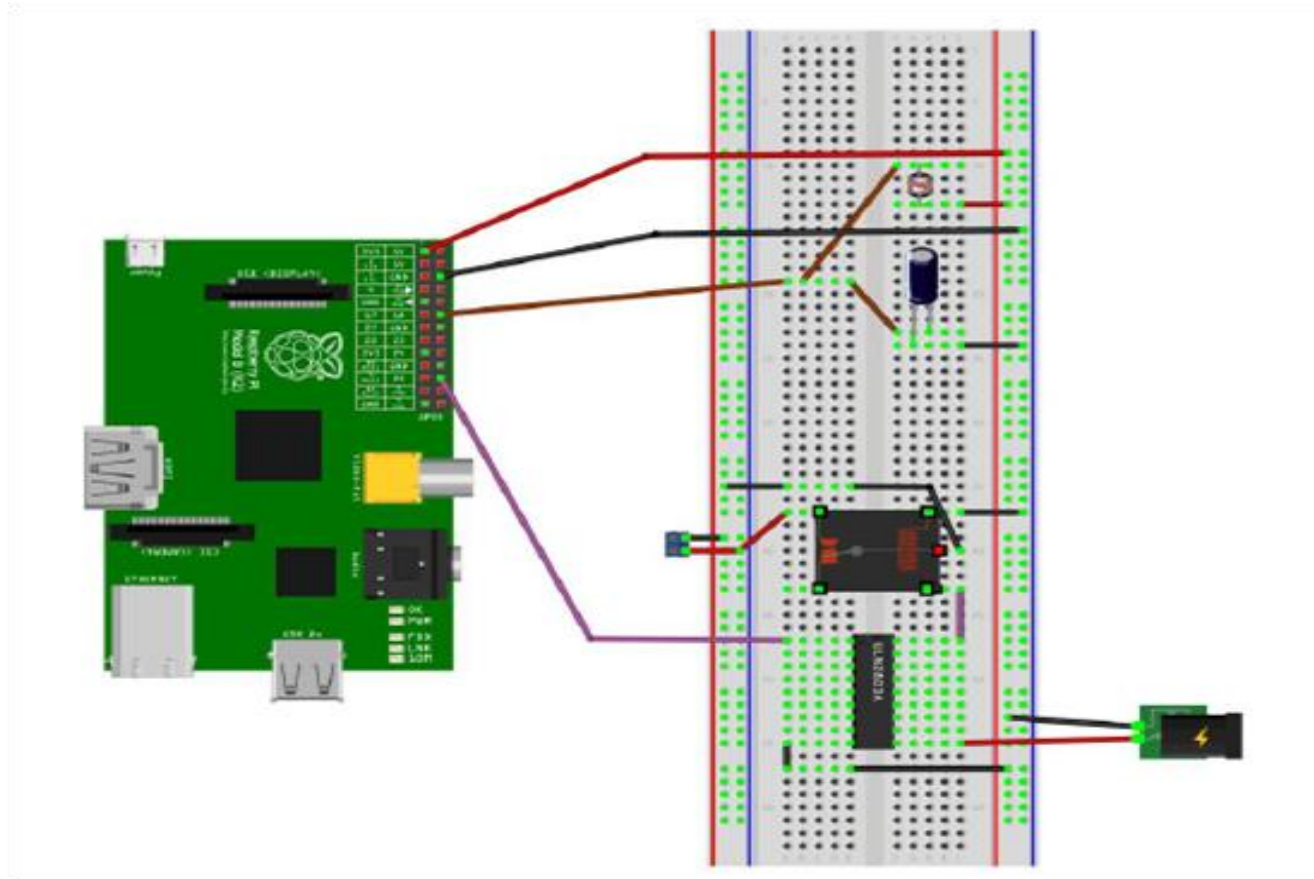
8. Operational View Specification

In this step, various options pertaining to the IoT system deployment and operation are defined, such as:

- ▶ Service hosting options
- ▶ Storage options
- ▶ Device options
- ▶ Application hosting options, etc.



9. Device & Component Integration



Step 10: Application Development

- Auto
 - Controls the light appliance automatically based on the lighting conditions in the room
- Light
 - When Auto mode is off, it is used for manually controlling the light appliance.
 - When Auto mode is on, it reflects the current state of the light appliance.

